89. Most of the explosions in mines occur due to the mixing of:

(a) Hydrogen with oxygen
(b) Oxygen with acetylene
(c) Methane with air
(d) Carbon dioxide with ethane

Ans. (c) : Methane explosions are one of the major source of explosion in coal mines. It occurs due to the presence of build up of methane (a by product of coal), when it comes in contact with a heat source. When air contains 9.5% of , it reaches the perfect oxidation point and explodes. So, the correct answer is 'Methane and air'.

90. Match List I with List II and select the correct answer using the codes given below the lists:

List I	List II
A. German Silver	1. Tin
B. Solder	2. Nickel
C. Bleaching Powder	3. Sodium
D. Hypo	4. Chlorine

	A	B	C	D
(a)	1	2	4	3
(b)	2	1	3	4
(c)	1	2	3	4
(d)	2	1	4	3

Ans. (d) : Nickel silver, also known as German silver, is a copper alloy with nickel and often zinc. Solders are typically made from tin or lead or a combination of both in the ratio of 63:37 respectively. Calcium hypochlorite, also known as bleaching powder, is a chemical compound with formula $Ca(ClO)_2$. It is widely used for water treatment and as a bleaching agent. This chemical is considered to be relatively stable and has greater available chlorine than sodium hypochlorite (liquid bleach). Hypo solution is the abbreviation for sodium thiosulphate or sodium hyposulphite, a chemical used to fix the image on photographic film after it has been developed.

91. Which one of the following materials is very hard and very ductile?

(a) Carborundum (b) Tungsten
(c) Cast iron (d) Nichrome

Ans. (d) : Nichrome is a non-magnetic alloy of nickel, chromium, and iron, usually used as a resistance wire. A common alloy is 80% nickel and 20% chromium, by mass. This alloying providenichrome properties like hardness and ductility.

BIOLOGY

92. Hybridoma technology is a new biotechnological approach for commercial production of:

(a) Monoclonal antibodies
(b) Interferon
(c) Antibiotics
(d) Alcohol

Ans. (a) : Hybridoma technology is being used for commercial production of monoclonal antibodies. The monoclonal antibodies are protective protein produced by the clone of single immune cell.

93. The sensation of fatigue in the muscles after prolonged strenuous physical work is caused by:

(a) A decrease in the supply of oxygen
(b) Minor wear and tear of muscles fibres
(c) The depletion of glucose
(d) The accumulation of lactic acid

Ans. (d) : After prolonged rigorous physical work, the body experiences fatigue due to accumulation of lactic acid in the muscles.

94. A small pouch containing, silica gel is often found in bottles of medicine in tablet or powder form because silica gel:

(a) Kills bacteria
(b) Kills germs and spores
(c) Absorbs moisture
(d) Absorbs all gases present inside the bottle

Ans. (c) : Silica gel granule is a network of inter-connecting microscopic pores, which attract and hold moisture by the phenomena known as absorption.

95. Canola refers to special type of oil seed mustard varieties bred for human consumption. The main characteristic of these varieties is that the:

(a) Seeds have very high oil content
(b) Oil is rich in unsaturated fatty acids
(c) Oil has long shelf-life
(d) Oil has very low Erucic acid content

Ans. (b) : Canola oil is the lowest in saturated fat, it is high in certain omega 3 fatty acid and mono saturated fats, which are found in nuts and are healthy for the heart.

96. Assertion A : In human beings, the females play a major role in determining the sex of the offspring.
Reason R : Women have two 'X' chromosomes.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (d) : Assertion is false, but R is true. In human being, male plays the major role in determining the sex of the offspring because male has XY-chromosome, while the female has two X-chromosome. XY combination of chromosome produces male child while XX combination produces female child.

97. Assertion A : "DNA Finger printing" has become a powerful tool to establish paternity and identity of criminals in rape and assault cases.
Reason R : Trace evidences such as hairs, saliva and dried semen are adequate for DNA analysis.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : DNA fingerprinting is a laboratory technique used to establish a link between biological evidence and a suspect in a criminal investigation. A DNA sample taken from a crime scene is compared with a DNA sample from a suspect. If the two DNA profiles are matched, then the suspect is proved culprit. For rape and assault cases, body fluid identification of stains such as differentiation of semen, saliva, vaginal fluid and menstrual blood can be critical in identifying probative evidence and corroborating the events of an assault.

98. Assertion A : In a motion picture, usually 24 frames are projected every second over the whole length of the film.
Reason R : An image formed on the retina of eye persists for about 0.1s after the removal of stimulus.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (c) : In a motion picture, usually 24 frames are projected every second over the whole length of the film because it was determined to be the minimum speed needed to capture video while still maintaining realistic motion. Hence A is true. The image of an object seen by our eyes persists on retina for about 1/16th of a second even after object has disappeared from our view. Hence R is false.

99. The 'stones' formed in human kidney consist mostly of

(a) Calcium oxalate (b) Sodium acetate
(c) Magnesium sulphate (d) Calcium

Ans. (a) : Stones are typically classified by their location: nephrolithiasis (in the kidney), ureterolithiasis (in the ureter), cystolithiasis (in the bladder), or by what they are made of (calcium oxalate, uric acid, struvite, cystine).

100. At which stage in its life cycle does the silk worm yield the fibre of commerce?

(a) Egg (b) Larva
(c) Pupa (d) Imago

Ans. (c) : Silk worm yield the fibre from pupa stage, which is commercially used to produce clothes.

101. Match List I (Diseases) with List II (Types of disease) and select the correct answer using the codes given below the lists:

List I	List II
A. Haemophilia	1. Deficiency disease
B. Diabetes	2. Genetic disease
C. Rickets disorder	3. Hormonal
D. Ringworm	4. Fungal infection

	A	B	C	D
(a)	2	3	4	1
(b)	2	3	1	4
(c)	3	2	1	4
(d)	3	2	4	1

Ans. (b) : Ringworm is a fungal infection in humans. Rickets is caused by deficiency of Vitamin D, thus a deficiency disease. Hemophilia is a genetic disorder and Diabetes is hormonal disorder that is caused by irregularities of Insulin, a vital hormone secreted by pancreas.

102. The blood glucose level is commonly expressed as:

(a) mm. of HG
(b) Milligram per deciliter
(c) Parts per million
(d) Grams per litre

Ans. (b) : The level of blood glucose level is measured by insulin and glucagon secreted by the pancreas. The amount of glucose is measured in mg/dL. A normal fasting (no food for eight hours) blood sugar level is between 70 and 99 mg/dL. A normal blood sugar level two hours after eating is less than 140 mg/dL.

103. Match List I (Endocrine glands) with List II (Hormones secreted) and select the correct answer:

List I	List II
A. Gonads	1. Insulin
B. Pituitary	2. Progesterone
C. Pancreas	3. Growth hormones
D. Adrenal	4. Cortisone